

Reading Notes for the Closed-Loop Hessian Failure Figures

Extension study based on arXiv:2606.05060v1

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Abstract

These notes explain the nine figures generated in the extension study of low-rank-Hessian closed-loop calibration for a neutral-atom controlled-Z (CZ) gate. The aim is not only to say whether a run succeeds, but to distinguish four possible causes of failure: a distortion that is too large for the quadratic model, rotation of the sensitive control subspace, measurement/fit noise, and a new physical error channel that is absent from the nominal Hessian.

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1 Minimal background

1.1 Complete symbol and label dictionary

This section defines the notation used in the figures and in the explanations below. In particular, Ω_c and Ω are different objects: $\Omega_c(t)$ is the full complex, time-dependent control, whereas Ω is its reference magnitude and the frequency unit used on several axes.

1.1.1 Pulse, time, and quantum states

Symbol or label	Meaning
t	Physical time during one gate pulse.
T	Total gate duration. Therefore t/T is normalized time, running from 0 at the pulse start to 1 at the pulse end.
$\Omega_c(t)$	Complex control Rabi frequency applied to the atoms. Its real and imaginary parts are the two Cartesian control quadratures.
Ω	Positive reference Rabi-frequency magnitude. For the analytical pulse $ \Omega_c(t) = \Omega$. It also sets the units of ω , detuning, and Hamiltonian-perturbation strength.
$ \Omega_c /\Omega$	Pulse magnitude divided by the reference magnitude. A value of 1 means no amplitude modulation.
$\phi(t)$	Time-dependent phase of the complex control: $\Omega_c(t) = \Omega_c(t) e^{i\phi(t)}$.
A	Dimensionless amplitude of the phase modulation.
ω	Angular frequency of the phase modulation, measured here in units of Ω .
ϕ_0	Phase offset inside the cosine modulation.
δ_0	Optional linear phase-ramp coefficient in $\phi(t) = A \cos(\omega t - \phi_0) + \delta_0 t$. It is set to zero in this run.
i	Imaginary unit, $i^2 = -1$. In a subscript such as λ_i or v_i , i instead denotes the mode number.
π	Circle constant. A phase of π radians is 180° .
$ 0\rangle, 1\rangle$	The two computational states of one atom.
$ r\rangle$	The Rydberg excited state of one atom.
$ 01\rangle, 10\rangle, 11\rangle$	Two-atom computational basis states. For example, $ 01\rangle = 0\rangle \otimes 1\rangle$.
$ 0r\rangle, r0\rangle$	Single-Rydberg leakage partners of $ 01\rangle$ and $ 10\rangle$.
$ W\rangle$	Symmetric blockaded Rydberg state $(r1\rangle + 1r\rangle)/\sqrt{2}$ coupled to $ 11\rangle$.
$\langle a b\rangle$	Inner product of states $ a\rangle$ and $ b\rangle$. An expression $ a\rangle\langle b $ maps $ b\rangle$ to $ a\rangle$.
h.c.	Hermitian conjugate; it adds the reverse couplings required to make the Hamiltonian Hermitian.
CZ	Controlled- Z gate: ideally only $ 11\rangle$ acquires an additional phase of π , up to correctable single-qubit Z phases.

1.1.2 Waveform distortion and Hessian geometry

Symbol or label	Meaning
$\Omega_0(t)$	Nominal analytical control waveform before an artificial distortion is added.
$\delta\Omega(t)$	Complex waveform distortion: $\delta\Omega(t) = \Omega_c(t) - \Omega_0(t)$.
$\delta\Omega$ without (t)	The real control vector obtained by stacking $\text{Re } \delta\Omega$ and $\text{Im } \delta\Omega$ at all time bins. Its length is $2N$.
$\text{Re}(z), \text{Im}(z)$	Real and imaginary parts of a complex quantity z .
$ z $	Absolute value for a real number or magnitude for a complex number.
$\ x\ $	Euclidean norm of a vector x ; $\ x\ ^2$ is its total squared amplitude.
N	Number of time bins, $N = 256$ in this run.
η	Relative root-mean-square waveform distortion, $\eta = \sqrt{\int \delta\Omega ^2 dt / \int \Omega_0 ^2 dt}$. It measures waveform size, not gate infidelity.
$\mathcal{H}_F$	Leading fidelity Hessian in the real waveform coordinates. It is the quadratic sensitivity matrix of the five first-order error channels.
$\delta\Omega^\top$	Transpose of the real distortion vector. Hence $\delta\Omega^\top \mathcal{H}_F \delta\Omega / 2$ is a scalar predicted infidelity.
$O(\ \delta\Omega\ ^3)$	Terms cubic and higher in distortion magnitude that are discarded by the leading quadratic approximation.
λ_i	i th Hessian eigenvalue, ordered from largest to smallest. A larger value means stronger local sensitivity along that eigenvector.
λ_1	Largest Hessian eigenvalue, used to normalize the spectrum.
$ \lambda_i /\lambda_1$	Relative magnitude of Hessian mode i . The declared active-rank threshold is 10^{-8} .
v_i	Normalized eigenvector associated with λ_i ; it is one waveform direction scanned by the closed loop.
$\text{Re}(v_i), \text{Im}(v_i)$	Real and imaginary control quadratures of eigenvector v_i .
$\sqrt{N} v_i$	Plotting rescaling of a unit-norm discrete eigenvector. Multiplication by $\sqrt{N}$ makes typical curve values order one; it does not change the direction.
$P_{\mathcal{H}_F}$	Orthogonal projector onto the five-dimensional space spanned by $v_1, \dots, v_5$.
$p_{\parallel}$	Fraction of distortion power in that five-dimensional space: $p_{\parallel} = \ P_{\mathcal{H}_F} \delta\Omega\ ^2 / \ \delta\Omega\ ^2$.
principal space	Span of the five active nominal Hessian eigenvectors.
null space	All nominal waveform directions orthogonal to that principal space; their leading quadratic prediction is zero.
principal angle	Standard angle comparing two subspaces. Zero degrees means alignment; 90 degrees means that at least one direction is orthogonal.
singular overlap	Singular value of the overlap matrix between two orthonormal subspace bases. One means perfect retention and zero means a lost direction.

1.1.3 Fidelity, leakage, phase, and noise

Symbol or label	Meaning
F	Average CZ gate fidelity after removing freely correctable local single-qubit Z phases. An ideal gate has $F = 1$.
$1 - F$	Gate infidelity. Smaller is better; success is declared when $1 - F \leq 10^{-5}$.
10^{-5}	The chosen success threshold, corresponding to fidelity $F \geq 0.99999$.
$P_{\text{leak},01}$	Probability that population starting in the $ 01\rangle$ sector finishes outside its computational state.
$P_{\text{leak},11}$	Corresponding leakage probability for the $ 11\rangle$ blockaded sector.
a_{01}, a_{10}, a_{11}	Complex return amplitudes of the three nontrivial computational sectors after the pulse. Their magnitudes encode survival and their arguments encode phase.
$\arg(z)$	Complex phase (argument) of z , expressed in radians.
θ_{CZ}	Nonlinear two-qubit phase $\arg(a_{11}) - \arg(a_{01}) - \arg(a_{10})$; the CZ target is π modulo 2π .
$\delta\theta$	Wrapped error of θ_{CZ} relative to π .
$\delta\theta^2/5$	Nonlinear-phase proxy used for the green bars in Figure 6. It is a diagnostic scale, not an independently additive probability.
σ_F	Standard deviation of independent additive Gaussian noise placed on each scalar fidelity evaluation in Figure 7.
non-positive fit	A seven-point quadratic line fit whose fitted curvature is zero or negative; this signals an unresolved or non-convex local minimum.
channel diagnostic	A leakage probability or phase-error proxy plotted to show which physical error remains.

1.1.4 Algorithmic labels appearing on the plots

Plot label	Meaning
mode	One normalized Hessian eigenvector v_i . There are five scanned modes.
accepted mode update	One completed one-dimensional scan followed by the applied coefficient update along one mode. Five updates make one complete cycle unless the success target stops the run early.
seven-point fit	Seven symmetric coefficient samples along one mode, followed by a bounded quadratic fit and an update to its fitted minimum.
exact line minimum	A simulation-only oracle that numerically minimizes the true propagated infidelity along one mode.
exact propagation	Direct unitary propagation of the stated Hamiltonian, without truncating the time dynamics to the Hessian approximation.
initial/final	Value before the first closed-loop update and after the last accepted update, respectively.
before loop/after loop	Same meaning as initial/final in Figure 9.
success fraction	Number of random seeds reaching the target divided by ten trials in that scan cell.
seed	Integer identifying one reproducible random smooth distortion shape.

Plot label	Meaning
cycle	One pass through the five nominal Hessian directions.
residual floor	Final infidelity remains above 10^{-5} after the allowed eight cycles.
perturbation strength/ Ω	Hamiltonian-error amplitude divided by the reference Rabi frequency; for example, 0.1 means 0.1Ω .
degrees	Unit of the principal angle; $180^\circ = \pi$ radians.
rank threshold	Relative-eigenvalue cutoff $ \lambda_i /\lambda_1 > 10^{-8}$ used to count active Hessian modes.
logarithmic axis	Equal vertical distances represent equal multiplicative ratios. Moving by one decade changes a value by a factor of ten.
median	Middle value of the ten seeds after sorting; it is less sensitive to an extreme seed than the arithmetic mean.
90th-percentile shading	In Figure 4, the shaded region runs from the median to a value exceeded by only about 10% of seeds.
relative model discrepancy	In Figure 4, $ E_{\text{nonlinear}} - E_{\text{quad}} /\max(E_{\text{nonlinear}}, 10^{-10})$.

1.2 Model and fidelity target

The calculation uses the perfect-blockade, symmetric two-atom model

$$H_{\text{gate}}(t) = \frac{1}{2} \left\{ \Omega_c(t) \left[|0r\rangle\langle 01| + |r0\rangle\langle 10| + \sqrt{2} |W\rangle\langle 11| \right] + \text{h.c.} \right\}. \quad (1)$$

The analytical control has constant magnitude and time-dependent phase,

$$\Omega_c(t) = \Omega e^{i\phi(t)}, \quad \phi(t) = A \cos(\omega t - \phi_0) + \delta_0 t, \quad (2)$$

with the published rounded parameters $A = 2\pi(0.1122)$, $\omega = 1.0431\Omega$, $\phi_0 = -0.7318$, and $\delta_0 = 0$, with $\Omega T/(2\pi) = 1.215$.

The plotted error is the average gate infidelity $1 - F$. A calibration trial is called successful if

$$1 - F \leq 10^{-5} \quad (3)$$

within eight complete cycles. One cycle scans all five nominal Hessian directions.

1.3 Why there are five directions

Close to an ideal pulse, the gate error is approximated by

$$1 - F = \frac{1}{2} \delta\Omega^\top \mathcal{H}_F \delta\Omega + O(\|\delta\Omega\|^3). \quad (4)$$

In this model there are two complex leakage amplitudes and one real nonlinear controlled phase. These give $2 + 2 + 1 = 5$ real first-order error channels, so the leading Hessian has rank five even though the discretized waveform has 512 real control coordinates.

The closed loop keeps these five directions fixed at their values for the nominal pulse. Along each direction it evaluates seven symmetric points, fits a quadratic, moves to the fitted minimum, and then continues to the next direction.

1.4 The two scan parameters

The relative waveform distortion is

$$\eta = \sqrt{\frac{\int_0^T |\delta\Omega(t)|^2 dt}{\int_0^T |\Omega_0(t)|^2 dt}}. \quad (5)$$

Thus $\eta = 0.1$ means a 10% root-mean-square waveform error, not a 10% gate infidelity.

The second parameter is the fraction of distortion power initially placed in the nominal five-dimensional principal space,

$$p_{\parallel} = \frac{\|P_{\mathcal{H}_F}\delta\Omega\|^2}{\|\delta\Omega\|^2}. \quad (6)$$

Here $p_{\parallel} = 1$ is a pure principal-space distortion, $p_{\parallel} = 0$ is a pure nominal-null-space distortion, and $p_{\parallel} = 0.5$ is mixed.

1.5 Executive summary

Question	Answer from this exploratory calculation
When does magnitude matter?	Principal-rich distortions first become unreliable between $\eta = 0.35$ and 0.60 .
What geometric signal accompanies failure?	The local five-dimensional sensitive subspace rotates by roughly 58° – 63° at $\eta = 0.60$ for principal-rich cases.
When does fit noise matter?	The transition lies between scalar fidelity noise $\sigma_F = 10^{-4}$ and 10^{-3} in the chosen seven-point estimator.
What is the most serious pathology?	A new leakage or atom-asymmetry channel produces a residual floor even when its raw perturbation is smaller than some correctable errors.
Are all “weird” waveforms failures?	No. A zero crossing, clipping, slow drift, and pure-null error were all corrected in the particular probes tested here.

2 Figure-by-figure explanation

2.1 Figure 1: the pulse and its five sensitive directions

How to read it. The top-left panel shows that the pulse magnitude is constant: $|\Omega_c(t)|/\Omega = 1$. The top-middle panel shows the phase modulation that actually shapes the gate. The top-right panel orders the Hessian eigenvalues from largest to smallest and divides them by the largest eigenvalue. Exactly five eigenvalues lie above the dashed rank threshold.

The lower panels show the real and imaginary quadratures of the five eigenvectors. Each curve is a direction in which the waveform can be changed. The optimizer scans coefficients along these shapes; it does not independently scan all 512 time-bin coordinates.

What it means. The rounded analytical pulse gives $1 - F = 6.60 \times 10^{-6}$ and a controlled-phase error of 1.94×10^{-4} rad. The ratio $|\lambda_6|/\lambda_1 = 2.68 \times 10^{-16}$ confirms the intended rank-five structure. This figure is the acceptance test for the rest of the study.

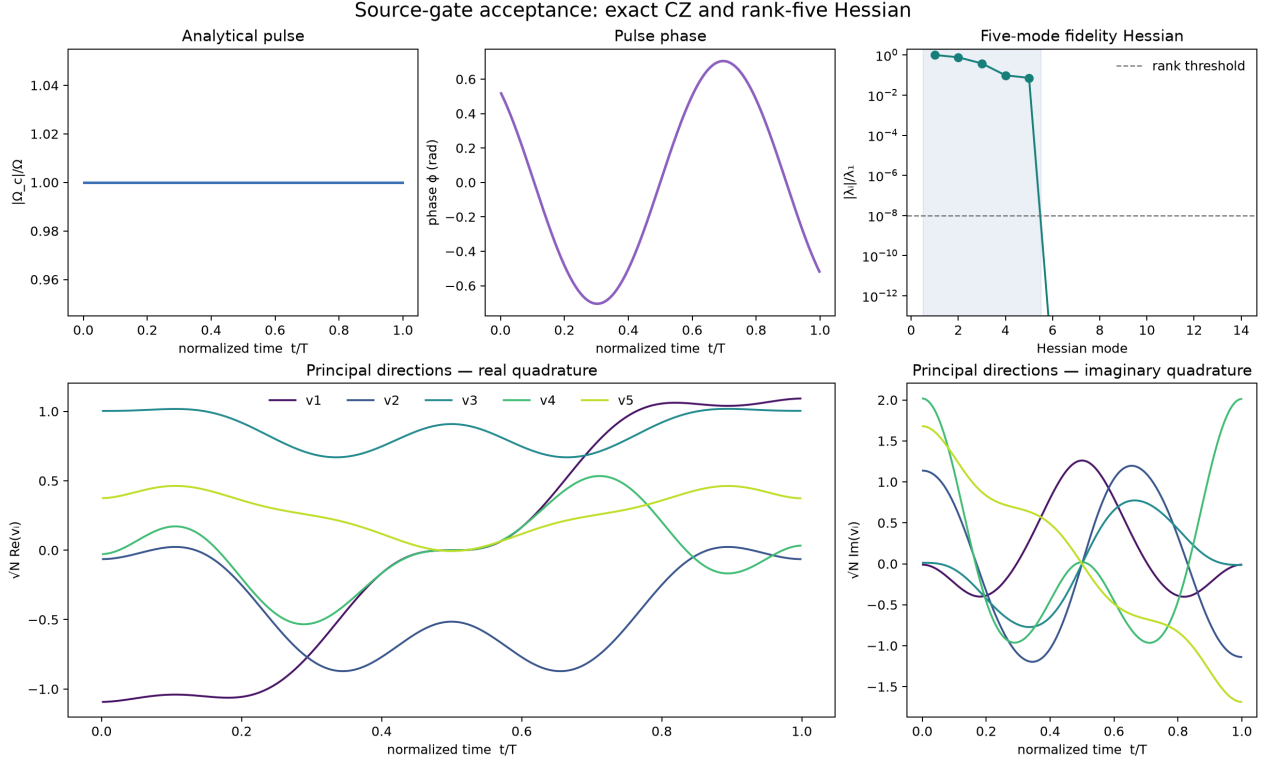


Figure 1: Nominal analytical pulse, leading Hessian spectrum, and the five principal waveform directions.

Important caution. “Exact CZ” in the figure title means exact numerical propagation of the specified model, not zero mathematical error. The published rounded pulse constants leave the small residual error stated above. Individual eigenvectors can also change sign without changing their physical meaning; only their span matters.

2.2 Figure 2: representative optimization trajectories

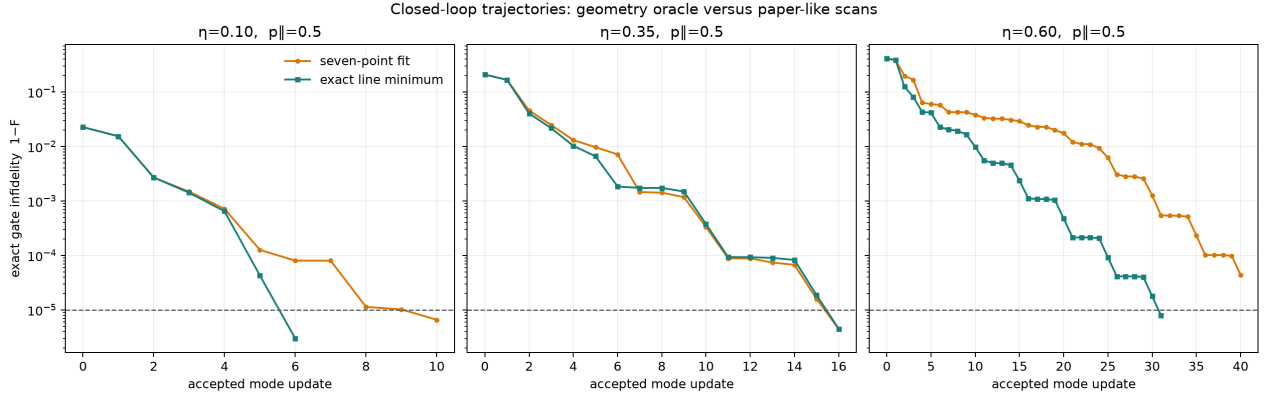


Figure 2: Gate infidelity after every accepted mode update for mixed distortions with $p_{||} = 0.5$.

How to read it. The vertical axis is logarithmic, and the dashed horizontal line is the success target $1 - F = 10^{-5}$. The horizontal axis counts accepted single-mode updates; five such updates are one complete cycle.

Orange circles use the experimentally plausible seven-point quadratic fit. Green squares use an exact one-dimensional minimizer that has access to the simulated fidelity function. The green result is an oracle diagnostic, not an experimentally available method.

What it means. At $\eta = 0.10$ and 0.35 , both methods converge below the target. At $\eta = 0.60$, the exact line minimizer still reaches $1 - F \simeq 7.8 \times 10^{-6}$, while the seven-point fit stops near 4.4×10^{-5} in this representative seed. Therefore this particular failure is not proof that the five directions contain no useful descent path. It is largely a failure of the local quadratic line fit on a non-parabolic landscape.

Important caution. This figure shows only one seed at each η . The population success fraction is in Figure 3. Flat sections are also not necessarily a total stall: a scanned direction may make little change before a later direction produces a large drop.

2.3 Figure 3: the closed-loop failure map

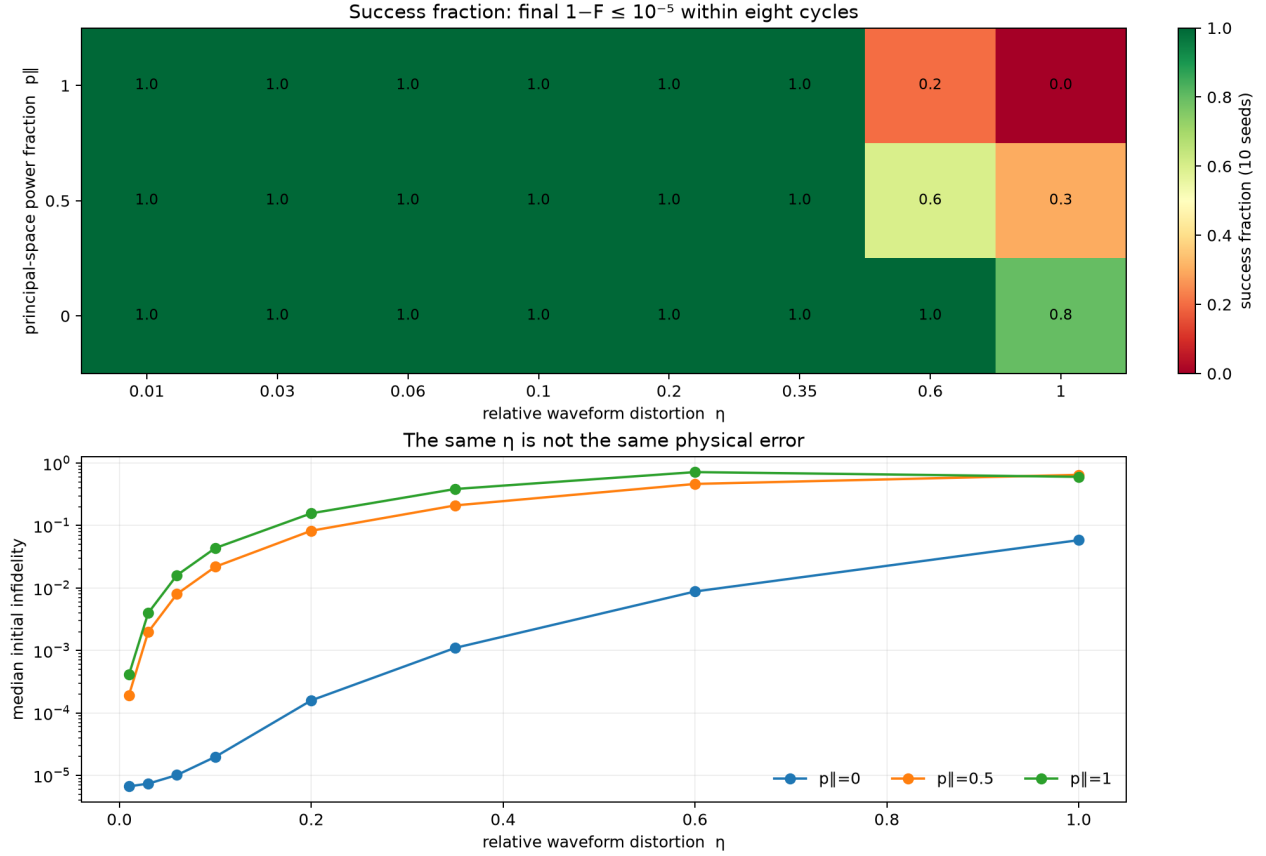


Figure 3: Success fraction over ten smooth random distortions per cell, with the corresponding median initial gate infidelity.

How to read it. In the upper heat map, green means that all ten seeds reached the target and red means that none did. Columns increase the waveform norm η ; rows increase the fraction $p_{||}$ initially placed in the five nominal Hessian directions.

The lower panel is essential: it shows the median gate error before optimization. At the same η , a principal-space distortion can cause orders of magnitude more gate error than a nominal-null-space distortion.

What it means. All three orientations succeed for all ten seeds through $\eta = 0.35$. At $\eta = 0.60$, success is

$$10/10 \quad (p_{||} = 0), \quad 6/10 \quad (p_{||} = 0.5), \quad 2/10 \quad (p_{||} = 1).$$

At $\eta = 1$, the corresponding counts are 8/10, 3/10, and 0/10. Thus the first observed principal-rich failure boundary is bracketed by $0.35 < \eta \leq 0.60$.

Important caution. It would be wrong to conclude that null-space errors are intrinsically easier at equal physical gate error. They look easier here at equal waveform norm because the nominal Hessian does not see them at first order, so their initial gate infidelity is much smaller. Ten seeds give only an exploratory boundary; for example, 6/10 has a wide 95% Wilson interval of approximately 0.31–0.83.

2.4 Figure 4: when the quadratic Hessian model stops being accurate

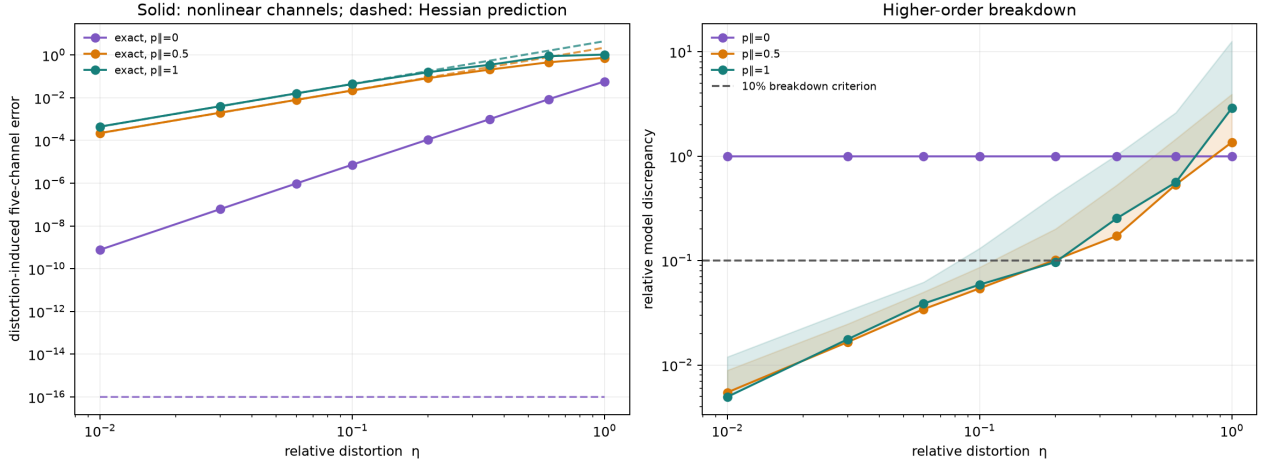


Figure 4: Nonlinear five-channel error compared with the leading Hessian prediction, and their relative discrepancy.

How to read it. In the left panel, solid curves are obtained by propagating the distorted pulse and evaluating the nonlinear leakage/phase channels. Dashed curves are the prediction

$$\frac{1}{2}\delta\Omega^T\mathcal{H}_F\delta\Omega.$$

The right panel plots their relative difference. The horizontal dashed line marks the chosen 10% breakdown criterion. Shading extends from the median to the 90th percentile across seeds.

What it means. For $p_{\parallel} = 0.5$, the median discrepancy is about 5.4% at $\eta = 0.10$, 10.1% at $\eta = 0.20$, and 53.4% at $\eta = 0.60$. Therefore the small-distortion expansion starts losing quantitative accuracy well before the success fraction collapses. The optimizer can tolerate some model error, but not unlimited model error.

Important caution. For $p_{\parallel} = 0$, the Hessian prediction is zero by construction. The exact higher-order channel error is nonzero, so the relative discrepancy is essentially one at every η . This does *not* mean a tiny null-space distortion is catastrophic; the left panel shows that its absolute error is very small at small η . Also, the plotted “exact” channel metric is baseline-relative and is not the raw total gate infidelity.

2.5 Figure 5: rotation of the locally sensitive subspace

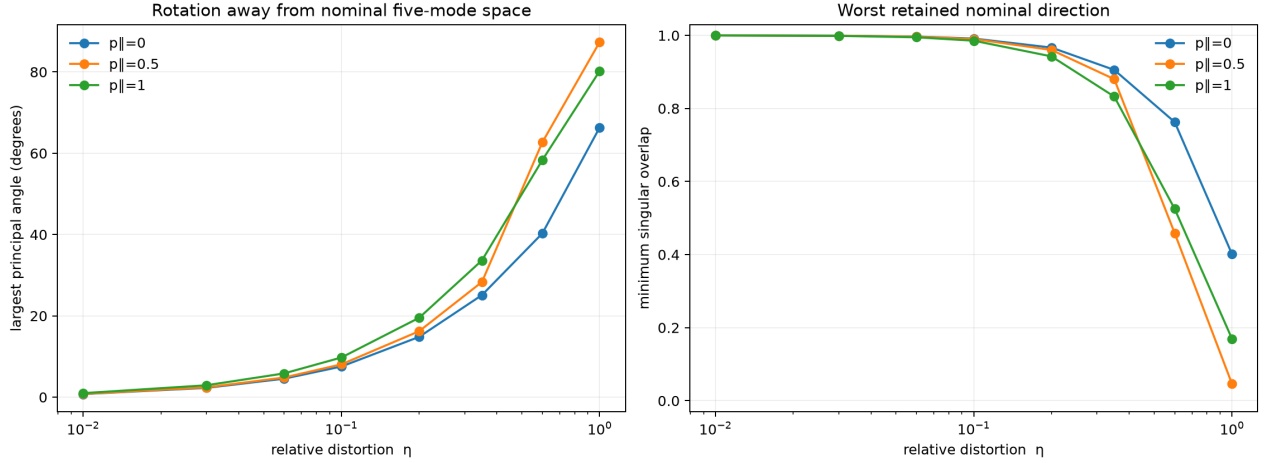


Figure 5: Geometric comparison of the nominal and distorted-pulse five-dimensional Hessian subspaces.

How to read it. At each distorted waveform, a new local five-mode Hessian is calculated. The left panel shows the largest principal angle between this local subspace and the nominal one. Zero degrees means perfect alignment; 90 degrees means that at least one local sensitive direction is orthogonal to its best nominal counterpart.

The right panel shows the smallest singular overlap. One means that every local direction is retained well; zero means that one useful direction has been almost completely lost by the fixed nominal scan space.

What it means. At $\eta = 0.35$, the largest angle is only about 25° – 34° , and every scan cell succeeds. At $\eta = 0.60$, it grows to about 40° – 63° and the principal-rich success fraction becomes seed-dependent. For the mixed case at $\eta = 1$, the angle is 87.4° and the worst overlap is only 0.046.

Important caution. The local Hessian still has five active modes; the issue is rotation, not a change from rank five to another rank. The correlation between rotation and failure is strong in this scan, but it is a diagnostic rather than a formal proof of causation.

2.6 Figure 6: which error channels remain after calibration?

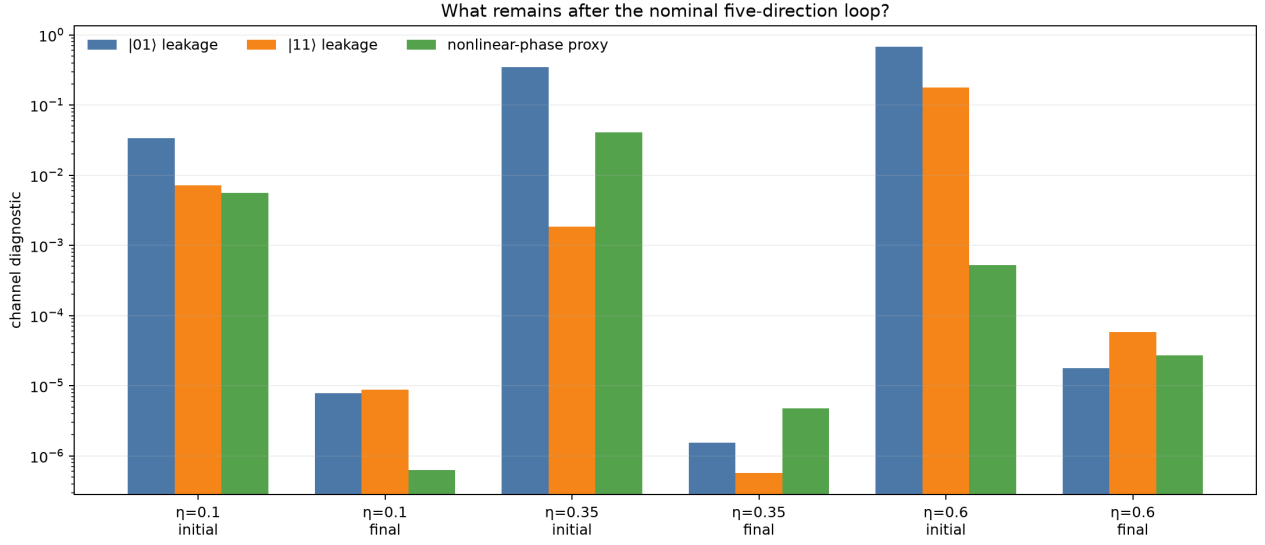


Figure 6: Initial and final leakage/phase diagnostics for one mixed seed at three distortion strengths.

How to read it. Each η has an initial group and a final group. Blue is leakage from the $|01\rangle$ sector, orange is leakage from the $|11\rangle$ sector, and green is a nonlinear-phase proxy $\delta\theta^2/5$. The vertical axis is logarithmic, so a one-decade drop is a factor of ten.

What it means. At $\eta = 0.10$ and 0.35 , the loop suppresses all three diagnostics to roughly the target scale or below. For the failed $\eta = 0.60$ trajectory, the final values are approximately

$$P_{\text{leak},01} = 1.79 \times 10^{-5}, \quad P_{\text{leak},11} = 5.85 \times 10^{-5}, \quad \delta\theta^2/5 = 2.71 \times 10^{-5}.$$

The plateau is therefore a mixture of leakage and phase error, not a single bad phase coefficient.

Important caution. These bars are channel diagnostics, not three exactly additive pieces of the total infidelity. Their fidelity weights differ, and higher-order cross terms remain. Compare orders of magnitude and before/after suppression, not the sum of bar heights.

2.7 Figure 7: failure caused by noisy fidelity estimates

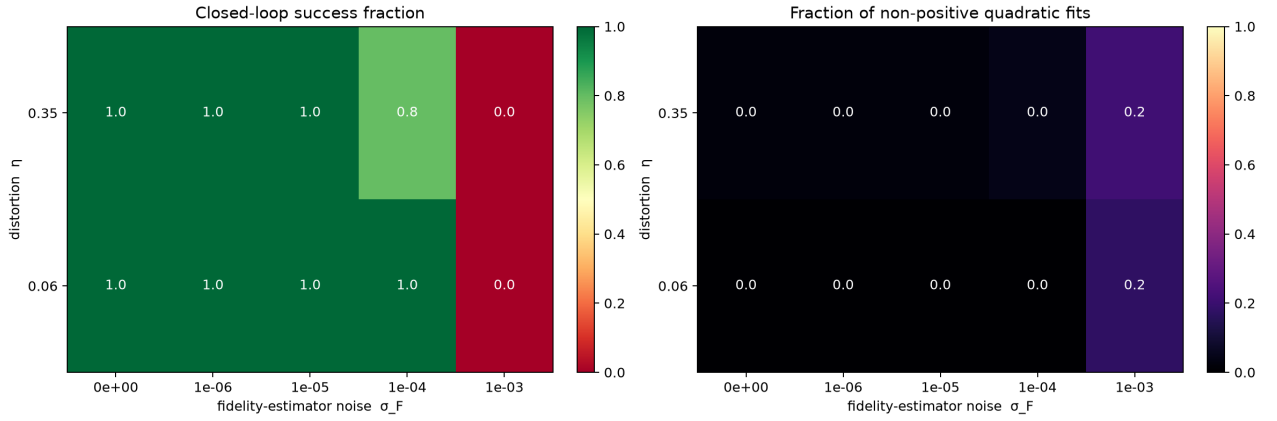


Figure 7: Success and quadratic-fit stability versus additive scalar fidelity-estimator noise.

How to read it. The left heat map is the success fraction over ten seeds. The right heat map is the median fraction of line scans whose fitted quadratic curvature is non-positive. A non-positive fitted curvature is a warning that noise has overwhelmed the local convex signal or that the sampled line is strongly non-quadratic.

What it means. For $\eta = 0.06$, all seeds succeed through $\sigma_F = 10^{-4}$ and none succeeds at 10^{-3} . For $\eta = 0.35$, success is 8/10 at 10^{-4} and 0/10 at 10^{-3} . Thus this estimator's noise boundary lies between 10^{-4} and 10^{-3} in absolute fidelity units.

Important caution. The noise is independent additive Gaussian noise on a scalar fidelity estimate. It is not a shot-resolved randomized-benchmarking or survival probability model. The threshold therefore diagnoses numerical conditioning of this seven-point rule; it is not a universal experimental shot-count requirement. The displayed value 0.2 at $\sigma_F = 10^{-3}$ is rounded from medians near 0.18–0.21.

2.8 Figure 8: six “weird” or ill-conditioned controls

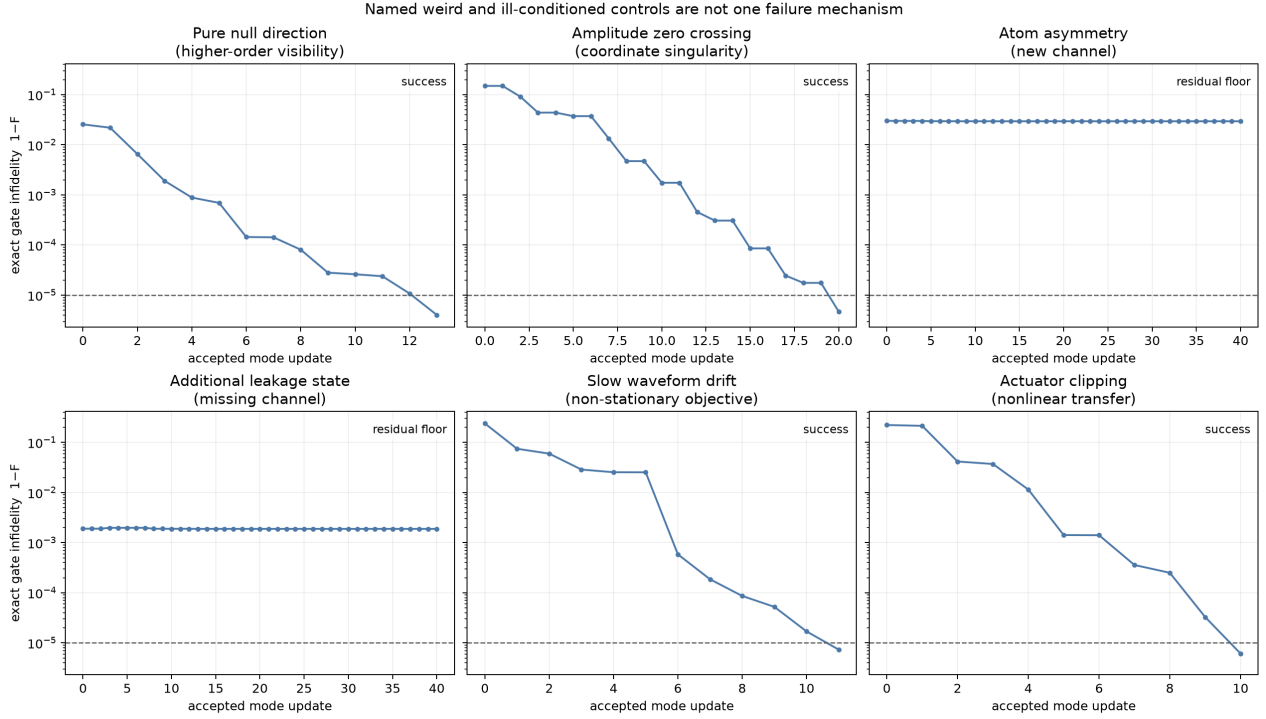


Figure 8: Infidelity trajectories for six explicitly defined stress tests.

How to read it. Every panel uses the same logarithmic infidelity axis and the same dashed success target. “Residual floor” means that eight cycles did not cross the target; “success” means that the trajectory did.

What it means.

- **Pure null direction:** a large $\eta = 0.60$ distortion that is invisible to the nominal Hessian at first order still converges to 4.04×10^{-6} because higher-order dynamics make it visible.
- **Amplitude zero crossing:** a localized 105% cancellation of the complex pulse converges to 4.72×10^{-6} . Cartesian quadratures remain regular when the amplitude crosses zero.
- **Atom asymmetry:** opposite differential drives on the two single-excitation sectors leave a floor near 2.95×10^{-2} .
- **Additional leakage state:** coupling one sector to a new state leaves a floor near 1.88×10^{-3} .
- **Slow waveform drift:** a null-like drift of norm $\eta = 0.02$ per cycle is tracked successfully, reaching 7.35×10^{-6} .
- **Actuator clipping:** a nonlinear amplitude cap at 1.05Ω is also compensated in this case, reaching 6.12×10^{-6} .

Important caution. These are single representative probes, not probability distributions. The conclusion is not that drift and clipping are always safe. It is that “weird-looking” control coordinates alone do not determine failure. The two clear floors arise from missing physical channels that the fixed five-direction space was never designed to control.

2.9 Figure 9: in-span detuning versus a new leakage channel

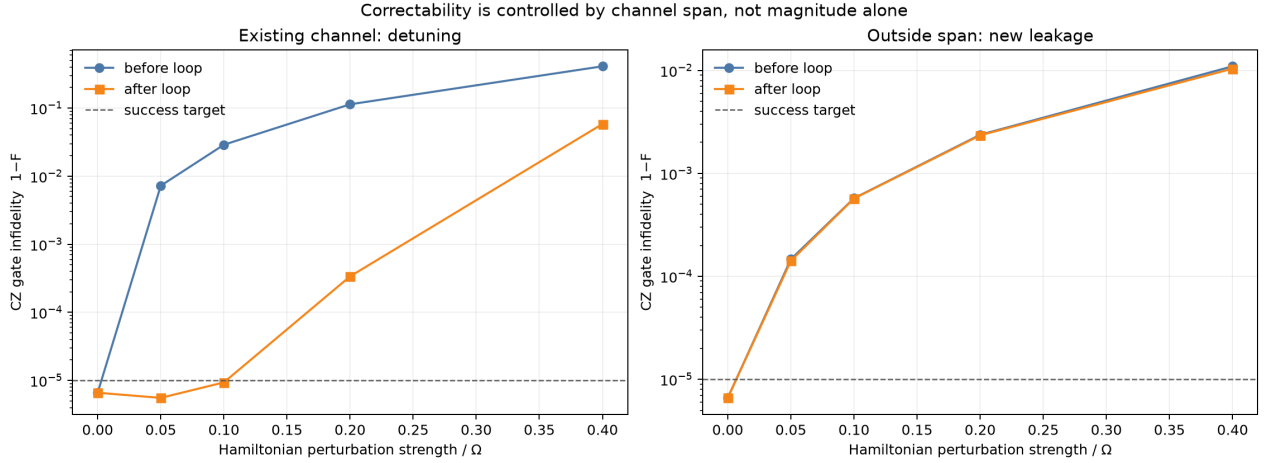


Figure 9: Gate infidelity before and after the loop for two Hamiltonian-error families.

How to read it. Blue circles show the gate before calibration and orange squares show it after calibration. The horizontal dashed line is the target. The left panel adds a detuning term that modifies the already existing dynamical channels. The right panel couples the system to an additional leakage state, creating a response absent from the nominal five-channel model.

What it means. A detuning of 0.10Ω is corrected from $1 - F = 2.89 \times 10^{-2}$ to 9.35×10^{-6} . At 0.20Ω , the loop still improves the gate strongly but stops at 3.37×10^{-4} .

By contrast, a new leakage coupling of only 0.05Ω starts with a much smaller error but remains at 1.41×10^{-4} after calibration. This is the central physical message: correctability depends on whether the error lies in the response span of the calibrated directions, not only on the raw perturbation magnitude.

Important caution. “Existing channel” does not mean that arbitrary detuning is always correctable. The fixed scan succeeds only through the tested 0.10Ω point under the strict 10^{-5} criterion. Correcting a new leakage channel would generally require measuring that channel and enlarging the Hessian/scan space.

3 How the figures fit together

The figures support the following diagnostic sequence:

1. Use Figure 1 to verify that the nominal gate and rank-five construction are correct.
2. Use Figure 3 to locate the empirical failure region.
3. Use Figure 4 to ask whether the small-distortion expansion has failed.
4. Use Figure 5 to ask whether the locally sensitive subspace has rotated away from the fixed nominal directions.
5. Use Figure 2 to distinguish a poor seven-point fit from the absence of a useful descent path.

6. Use Figure 6 to identify which nominal channels remain.
7. Use Figure 7 to test whether estimator noise makes line curvature unresolvable.
8. Use Figures 8 and 9 to test whether the device contains a genuinely new physical channel.

The practical lesson is that there is no single universal “maximum $\|\delta\Omega\|$ ”. A useful failure criterion must report at least the waveform norm, the initial physical infidelity, the orientation relative to the nominal Hessian, the rotation of the local subspace, estimator noise, and whether new Hamiltonian channels are present.

4 Scope and reproducibility

This is an exploratory perfect-blockade unitary study with ten random seeds per core scan cell. It does not include spontaneous decay, Doppler noise, finite blockade, or the full four-level experimental ytterbium model. The seven-point fit rule is an explicit extension choice because the paper does not publish all experimental scan-width and fitting details.

The paper is available at <https://arxiv.org/pdf/2606.05060v1>. The complete calculation is implemented in `tracks/qcs/solutions/hessian_loop_failure_map.py`. The numerical tables are in the run’s `data/` directory, and the figures are in `figs/`.